NLP Driven Agentic AI Systems for Software Engineering: A Survey in Resource Constrained Development Environments in Bangladesh

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Abstract

Imagine a world where intelligent agents can read one's mind, well, almost. Agentic AI is making that world real by thinking, planning and acting on its own, helping developers navigate the maze of software creation. Natural Language Processing (NLP) is the bridge that lets these agents understand instructions, explore documentation and assist in coding, debugging and more. This paper travels through the world of NLP based agentic AI systems, looking at how they work, the data they use and the tasks they tackle. But these systems often need constant internet, heavy computing power and struggle to adapt to changing work environments. Unlike other studies that only show blueprints, the study focuses on real world challenges, especially for freelancers and small companies in places like Bangladesh, where resources are limited. It also explores the path forward, creating lightweight, adaptable and internet smart Agentic AI tools that can actually thrive in the messy, unpredictable world of software development. By connecting AI's potential to practical realities, this work shows how intelligent agents can truly become partners for developers, not just powerful ideas on paper.

Keywords - **Agentic AI, Natural Language Processing, Software Engineering, Resource Constrained Environments, Freelance Development, SMEs, Intelligent Software Agents**
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1 Introduction

Artificial Intelligence (AI) is entering a new era in which machines are no longer limited to executing predefined instructions but are increasingly capable of understanding goals, making decisions and performing complex tasks. AI is not just a quiet tool sitting behind the screen, waiting for commands. It is now transforming into active agents that can think about problems, plan their next moves and carry out complex tasks with minimal human supervision. This transformation has given rise to the concept of Agentic AI, where AI systems function not merely as text generators but as intelligent agents capable of planning tasks and interacting with tools (Plaat et al., 2025).

In this paradigm, Agentic AI is about building systems that can act on their own. This usually involves multi agent systems (MAS), where specialized agents work together, communicate and coordinate to solve problems too big for any single agent (Sun et al., 2025). For example, an agentic system designed for automated bug fixing may include a bug analysis agent to examine error logs and identify root causes, a code navigation agent to locate the affected parts of the codebase, a patch generation agent to produce potential fixes, a testing agent to validate the correctness of the solution and a review agent to refine and improve the output. This coordinated workflow demonstrates how agentic systems can transform simple bug reports into reliable, executable software fixes.

To establish a clear conceptual foundation, it is important to distinguish between AI agents and Agentic AI systems. An AI agent refers to a computational entity that can perceive its environment, reason about possible actions and perform tasks in order to achieve a specific objective. In many recent systems, these agents are built on top of Large Language Models that provide reasoning, planning and communication capabilities (Zhao et al., 2024). Agentic AI, on the other hand, represents a broader architectural paradigm in which one or multiple agents collaborate to accomplish complex goals within a structured workflow (Xie et al., 2024).

Software engineering is fundamentally driven by language based interaction. Developers communicate requirements, describe bugs, review code, interpret documentation and coordinate tasks primarily through natural language. As a result, the integration of Natural Language Processing (NLP) (Mikolov et al., 2013) with agentic architectures has become a key enabler for intelligent software development systems. This language centric nature of software development makes Natural Language Processing (NLP) a critical interface for modern agentic systems. By interpreting requirements, generating code, summarizing documentation and coordinating tasks through language, NLP driven agents can function as collaborative assistants throughout the software development lifecycle.

Table 1 summarizes representative research works on agentic AI systems across different domains. As shown in the table:

- **Scattered Research:** Existing studies are spread across multiple domains, making it difficult to form a unified understanding of agentic system design and evaluation.
- **Limited Focus on Software Engineering:** Most research focuses on domain specific applications or architectures, with fewer studies analyzing support for coding, debugging, testing and deployment.
- **Persistent System Challenges:** Current systems still face issues such as high computational requirements, unreliable outputs and limited scalability.
- **Terminology Ambiguity:** Terms such as AI agents, LLM (Vaswani et al., 2017) based agents and multi agent systems are often used interchangeably, creating conceptual confusion (Brohi et al., 2025).
- **Failure Awareness:** Limited attention is given to detecting and recovering from hallucinations, incorrect code generation or tool misuse.

Beyond these research gaps, current surveys often overlook several practical deployment constraints that significantly influence the usability of agentic systems in real world environments.

- **Unstable Internet Connectivity:** Many systems assume continuous internet access, while real world connectivity can be slow or unstable.
- **Freelance Development Workflows:** Freelancers often handle vague client instructions, multiple projects and communication mainly through natural language.
- **Resource Limitations in Small Companies:** SMEs operate with limited computational resources, making large scale agentic systems difficult to adopt (Organisation for Economic Co-operation and Development, 2005).

These challenges are particularly significant in developing countries such as Bangladesh, where internet connectivity varies across regions, freelance software development plays a major role in the digital economy and many companies operate as small or medium sized enterprises. Consequently, designing **resource aware and connectivity resilient agentic systems** is essential for enabling practical deployment in such environments.

To address these limitations, this paper focuses on the role of NLP driven agentic AI systems in software engineering. The study carefully selects approximately 40 research papers from diverse and reputable academic sources in order to construct a meaningful and representative review Rather than restricting the selection to a single domain, the study deliberately embraces a cross domain perspective, drawing from research in agentic AI, natural language processing, multi agent systems and software engineering. The main contributions of this paper are summarized as follows:

- Provide a structured review of recent agentic AI systems with a particular focus on their application in software engineering tasks.
- Compare existing agentic systems across different domains and analyze their applicability to software engineering environments.
- Present a simplified taxonomy of agentic architectures based on their functional roles within the software development lifecycle (Pressman, 2010).

Table 1: Representative Research on Agentic AI Systems Across Domains

References	Focus	Key Contributions	Limitations
Yang et al. (2025)	Knowledge Graph	Introduces an agentic framework enabling LLMs to interact with graph structured knowledge.	Limited evaluation on large scale knowledge graphs.
Chauhan et al. (2026)	Healthcare	Proposes an adaptive agent-based system for symptom clarification and disease prediction.	Small dataset size and reliance on synthetic data.
Swati et al. (2026)	Agriculture	Develops a multi-agent framework for autonomous decision support in smart agriculture.	Performance depends on input data reliability.
Haldar et al. (2025)	Education	Automates short answer grading using agentic reasoning and RAG.	Difficulty handling highly subjective responses.
Chomphooyod et al. (2025)	Education / Syllabus	Multi-agentic AI for automatic course syllabus generation using LangGraph.	Requires high data consistency for accuracy.
Obeng et al. (2025)	Cybersecurity	Security framework for home AI to note cyber threats in preventive healthcare.	Vulnerable to specific prompt injection attacks.
Fahmi and Bousmah (2025)	Student Remediation	Agentic AI for student diagnosis and individualized remediation.	Scalability in diverse educational environments.
Fang et al. (2025)	Gaming / Games	A novel hierarchical agent framework for imperfect-information games.	Complexity in multi-agent strategy coordination.

- Introduce practical design considerations for real world deployment, including connectivity aware agent systems, support for freelance development workflows and lightweight architectures suitable for small and medium sized companies.

- Introduce the concept of **resource aware and connectivity resilient agentic systems** for software engineering environments in developing regions.

- Provide a context aware perspective by discussing how agentic AI systems can be adapted for developing regions such as Bangladesh, where resource constraints and freelance based development ecosystems are common.

The remainder of this paper is organized as follows. Section 2 provides the background and foundational concepts of Agentic AI and LLM based agents. Section 3 presents the proposed framework for context aware agent collaboration in software engineering. Section 4 provides a comparative analysis of agentic systems focusing on software related applications. Section 5 discusses key challenges in real world environments. Finally, Section 6 concludes the paper and outlines future research directions.

2 Background and Foundations of Agentic AI

2.1 Evolution of Agentic AI

The development of agentic AI can be seen as a natural continuation of the broader evolution of artificial intelligence. Imagine AI as a student in school:

- Early AI (rule-based systems) was like a student who could only follow exact instructions from the teacher. If asked "2 + 3," it answers correctly, but any variation confuses it.

- Machine learning AI became like a student who learns from examples. Even if a new problem appears, it can reason and guess the correct answer.

- Agentic AI is like a young scientist. It does not wait for instructions, it can set its own goals, plan steps, use tools and adapt to new challenges. For example, it could figure out how to build a small robot by reading manuals, testing ideas and improving its design iteratively.

In short, Agentic AI extends traditional AI by combining learning, reasoning and autonomous action, enabling systems to perform complex tasks with minimal human guidance which has been illustrated in Figure 1 (Russell and Norvig, 2021; Wang et al., 2023; Yao et al., 2023).

The emergence of Large Language Models (LLMs), particularly transformer based architectures, has significantly accelerated this transition (Brown et al., 2020; Wang et al., 2023). These models provide strong reasoning and language understanding capabilities, which, when combined with planning mechanisms, memory and external tools, enable the development of autonomous AI agents.

Recent research highlights a shift from static generative models to dynamic agentic architectures (Yao et al., 2023; Park et al., 2023). For example, agentic frameworks have been proposed for complex engineering tasks such as hardware code generation, where multiple agents collaborate to design and validate system components (Gadde et al., 2025a).

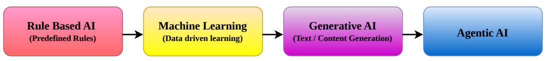

Figure 1: Evolution of Artificial Intelligence Toward Agentic Systems

In essence, agentic AI combines the learning ability of modern ML systems with planning, reasoning and tool usage, creating systems that are not only intelligent but also adaptive and capable of long term goal directed behavior. This evolution illustrated in Figure 1 marks a significant step toward autonomous artificial intelligence that can act effectively in real world environments.

2.2 Applications of Agentic AI

Table 2 summarizes key application areas and representative works of Agentic AI systems.

2.3 Taxonomy of Agentic AI Systems

A typical Agentic AI architecture includes several core components such as a planning module, memory system, reasoning engine and action execution layer. Based on existing literature and architectural design patterns, agentic AI systems can be broadly categorized into structural paradigms. Understanding these patterns is important for analyzing how agentic systems can be adapted to practical environments such as software engineering workflows. Figure 2 presents a high level taxonomy of agentic AI systems that organizes the current research landscape into three major categories:

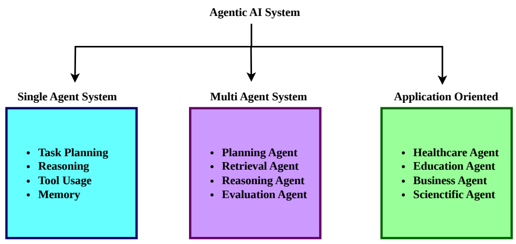

Figure 2: Taxonomy of Agentic AI Architectures

Single Agent Systems: A single autonomous agent handles task planning, reasoning and execution, and is effective for moderately complex tasks such as document analysis, code generation and information retrieval.

Multi Agent Systems: Multiple specialized agents collaborate, each responsible for a specific role, enabling the system to solve more complex and long term problems.

Application Oriented Agentic Systems: These systems apply agentic architectures to specific domains such as healthcare, education, optimization and software engineering, with this survey focusing on their role in software engineering workflows.

3 Agentic AI for Software Engineering: An NLP Centric and Resource Aware Perspective

Unlike many existing surveys that primarily focus on architectural designs of agentic systems, this paper examines agentic AI from an NLP driven software engineering perspective with particular attention to resource constrained development environments as depicted in Figure 3. In the world of software engineering, language is code, and understanding it intelligently is where NLP powered agentic AI becomes a game changer. Software engineering workflows are inherently language intensive, involving requirement descriptions, bug reports, documentation, code comments and developer discussions. As a result, Natural Language Processing becomes a critical interface through which agentic systems can understand, coordinate and assist development tasks.

Table 2: Application Domains of Agentic AI Systems

Domain	Application Task	Representative Papers
Healthcare	Symptom clarification, clinical decision support, medical report summarization, medical knowledge retrieval, severity detection	(Chauhan et al., 2026; Amadi and Ojo, 2025; Patel et al., 2025; Murali et al., 2025; He et al., 2026; Al-Bashrawi et al., 2026; Narang, 2025; Salomó-López et al., 2025)
Education	Short answer grading, personalized tutoring, student performance prediction	(Haldar et al., 2025; Hasan et al., 2026; George et al., 2025)
Agriculture	Crop monitoring and plant disease detection	(Swati et al., 2026; Giloni et al., 2025)
Software Engineering / Automation	Code generation, bug detection, programming assistance, software automation	(Dhiman et al., 2026; Karunanayake, 2025; Obeng et al., 2025)
Finance and Business Analytics	Consumer segmentation, banking analytics, strategic planning, risk prediction	(Chinnaraju, 2025; Chua et al., 2025; Maheshkar et al., 2026; Khamis, 2026)
Conversational and LLM Systems	Chatbots, dialogue systems, prompt optimization, LLM orchestration	(Honna et al., 2025; Madrid-García et al., 2025; Yerram et al., 2025; Moore and Tatonetti, 2025; Bai et al., 2025)
AI Research and Knowledge Systems	Literature review automation, document analysis, research assistant systems, conceptual AI studies	(Fang et al., 2025; Nooralahzadeh et al., 2025; Adawadkar, 2025)
Optimization and Multi Agent Systems	Multi agent coordination and decision optimization	(yuk; Pai, 2025)
Smart Systems (Cities, Robotics, Emergency)	Traffic optimization, autonomous planning, disaster response coordination	(Fahmi and Bousmah, 2025; Kanumolu et al., 2025; Aththanayake et al., 2025)
Hardware / Scientific Applications	Circuit design automation, graph learning, brain data analysis, multimodal reasoning	(Yang et al., 2025; Gadde et al., 2025b; Webb et al., 2025; Dabral et al., 2025)

Yet most agentic AI research assumes perfect conditions: stable internet, powerful machines and uninterrupted workflows. In practice, conditions are far from ideal. In countries like Bangladesh, freelance developers often contend with unstable connectivity, SMEs function with limited infrastructure and resources are frequently constrained. In such settings, AI must be smart, lightweight, adaptive. In the following, three deployment perspectives of NLP driven agentic systems have been explored in software engineering environments in Bangladesh.

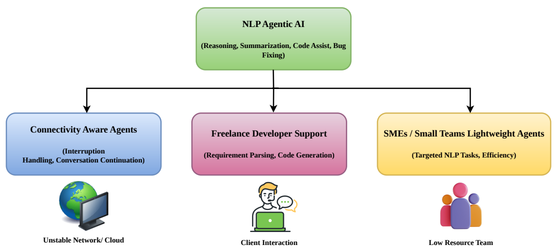

Figure 3: Resource Aware and Connectivity Resilient Agentic Systems

3.1 Connectivity Aware NLP Agents in Bangladesh

In many parts of Bangladesh, interacting with NLP driven intelligent systems is not always seamless. Modern software engineering workflows increasingly rely on natural language interactions, where developers describe requirements, discuss bugs and coordinate development tasks through text. As a result, NLP based agentic AI systems act as conversational interfaces that translate human instructions into executable software actions.

However, in real world environments such interactions are not always stable. Conversations with NLP based agentic systems may be interrupted by unstable networks, delayed responses or sudden disconnections. While the interaction pauses, the development workflow itself continues, creating challenges for language driven agents that rely on continuous dialogue.

Most existing agentic architectures assume persistent connectivity and real time access to large language models and external tools (Wang et al., 2023; Yao et al., 2023). Recent agentic NLP frameworks such as GraphAgent and RUBRA demonstrate how language models can reason over structured knowledge and retrieved context to support complex tasks (Yang et al., 2025; Haldar et al., 2025). However, these systems are typically designed for stable cloud environments, where uninterrupted communication between users and agents is assumed.

This creates a critical mismatch between system design and practical use. In connectivity limited environments, NLP agents should not only understand language but also continue conversations even if the connection is interrupted. These systems should be able to save the conversation, handle incomplete instructions and continue working when the internet is back.

In this situation, NLP is not just for understanding language. It also helps different AI agents work together smoothly, even during connection problems. This makes agentic AI more practical and useful, especially in countries like Bangladesh, where internet and system limitations are common.

3.2 NLP Agentic Support for Freelance Developers

For freelance developers, software engineering is not just about writing code, it is a continuous language driven interaction with clients. Most freelance projects begin with natural language, including client requirements, vague instructions, iterative feedback and evolving expectations. In this dynamic environment, understanding and acting on language becomes as important as technical execution.

NLP based agentic AI systems have the potential to transform this workflow by acting as intelligent intermediaries between human intent and technical implementation. Using large language models and reasoning frameworks, these agents can interpret client instructions, summarize requirements, translate them into structured development plans and generate corresponding code artifacts (Wang et al., 2023; Yao et al., 2023). In addition, recent agentic systems show that language based reasoning and tool use can help with complex development tasks such as debugging, writing documentation and improving code step by step (Park et al., 2023).

Despite this strong alignment, most existing studies focus on organizational software pipelines, where tasks are structured and collaboration occurs within teams. Freelance development ecosystems remain largely underexplored, even though they rely heavily on natural language communication between developers and clients.

From an NLP perspective, freelance workflows present a unique opportunity for agentic systems to operate as collaborative partners rather than simple automation tools. By maintaining conversational context, interpreting evolving requirements and assisting in planning and execution, NLP based agents can support developers throughout the entire project lifecycle. This highlights a practical application of agentic AI in real world development environments, particularly relevant to the growing freelance software economy in developing regions such as Bangladesh.

3.3 Lightweight NLP Agentic Systems for Bangladeshi SMEs

Small and medium sized enterprises (SMEs) are businesses that operate with limited workforce, budget and technical infrastructure. In Bangladesh, a large portion of software development organizations fall into this category, including small startup teams building web applications, local IT service providers and software vendors developing custom solutions for clients. For instance, a small development team working on an E-commerce platform may consist of only a few developers responsible for requirement analysis, coding, testing and client communication simultaneously.

In such environments, software engineering workflows are often fast paced and resource constrained. Developers must handle multiple responsibilities with minimal tooling support. Importantly, natural language interactions play a central role in these workflows, including requirement discussions, client instructions, bug reports and project documentation. From an NLP perspective, these language artifacts form the primary interface through which developers communicate, reason about problems and coordinate development tasks.

However, most existing NLP driven agentic AI systems are designed for large scale infrastructures that assume continuous cloud connectivity and substantial computational resources (Brown et al., 2020; Yao et al., 2023). Such assumptions limit the practical applicability of these systems in SME environments, where computing infrastructure, GPU resources and stable connectivity may be limited.

In this case, the goal is not to build big autonomous AI systems. Instead, the focus is to create simple and efficient NLP agents for small and medium software teams. This changes the goal of agentic AI, not just about making powerful models, but about making them useful in real life. The aim is to ensure that AI support is available even in environments with limited resources.

4 Comparative Analysis of Agentic AI Systems in Software Engineering

This section presents a comparative analysis of representative agentic AI systems, focusing on their applicability to software engineering tasks and their limitations in practical deployment scenarios.

The comparison in Table 3 shows that most NLP based agentic AI systems rely on multi agent architectures and assume stable internet and high computing power, limiting their practicality in resource constrained environments such as freelance work and SMEs. This highlights a gap between current designs and real world usability, emphasizing the need for resource aware NLP agentic systems for software engineering ecosystems.

5 Challenges in Deploying Agentic AI

Despite the rapid advancement of NLP based agentic AI systems, several critical challenges remain that limit their adoption in real world software engineering environments. These challenges shown in Figure 4 become even more noticeable in places like Bangladesh, where limited internet access, lower computing power and different development practices make it harder to use these systems effectively.

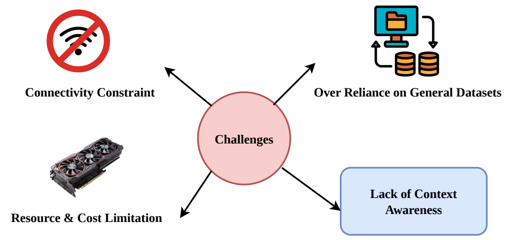

Figure 4: Challenges during Agentic AI Adoption

5.1 Connectivity Constraint

One of the primary challenges is the reliance on continuous internet connectivity. Most Agentic NLP systems are designed for always online environments, making them less effective in regions with unstable or intermittent network access. This creates a gap between system design and actual usage conditions.

5.2 Resource and Cost Limitation

Another major limitation is the high computational and financial cost associated with large scale NLP models. SMEs and freelance developers often lack the infrastructure required to deploy and maintain such systems, restricting their accessibility. Designing efficient NLP agents that operate with reduced computational overhead therefore becomes an important research direction.

5.3 Lack of Context Awareness

Existing systems often struggle to adapt to dynamic and unstructured software development environments. They typically operate well in predefined workflows but lack the flexibility to handle real time changes, informal communication and evolving requirements.

5.4 Over Reliance on General Datasets

Most agentic systems are trained on large, general purpose datasets, which may not reflect the specific needs of local software engineering contexts. This limits their effectiveness in handling region specific tasks and domain specific challenges. Developing localized datasets and domain specific corpora for

Table 3: Comparison of NLP Based Agentic AI Systems for Software Engineering

System	Architecture	Dataset Used	Application	Limitations
GraphAgent (Yang et al., 2025)	Multi Agent	Not specified / Knowledge Graph	Code + structured reasoning	Limited scalability
AI Software Eng. Agents (Maheshkar et al., 2026)	Multi Agent	GitHub repositories	Code generation, testing	High computational cost
Automated NLP Pipeline (Patel et al., 2025)	Single Agent	NLP datasets	Information extraction	Limited autonomy
Autonomous Workflow Agents (Richards and Yang, 2025)	Multi Agent	Workflow datasets	Task automation	Requires stable infrastructure
DouAgent (Fang et al., 2025)	Multi Agent	Simulation dataset	Task decomposition	Coordination complexity
Knowledge Retrieval Agents (George et al., 2025)	Single Agent	Knowledge base datasets	Documentation, retrieval	Limited reasoning depth
Data Analysis Agents (Nooralahzadeh et al., 2025)	Multi Agent	Analytical datasets	Data analysis	Resource intensive
RUBRA Framework (Haldar et al., 2025)	Single Agent (RAG based)	Retrieval datasets	Educational NLP + reasoning	Domain specific, not generalizable
Multi Agent System Framework (Sun et al., 2025)	Multi Agent	Simulation dataset	Distributed decision making	Generic design, lacks SE specialization

software engineering communication can significantly improve the effectiveness of NLP driven agentic systems.

6 Future Research Directions and Conclusion

Future research should focus on building lightweight and efficient NLP based agentic systems that can work reliably even under real world constraints such as unstable connectivity and limited computing resources. Exploring hybrid architectures, offline capabilities and task specific optimization techniques can make these systems more practical and accessible. In addition, developing local datasets and designing solutions tailored to freelance developers and SME based software environments can greatly increase the real world usefulness of agentic AI. Such efforts can help bridge the gap between theoretical research and practical software engineering applications.

This paper presents a focused survey on NLP based agentic AI systems in the context of software engineering, highlighting their capabilities, architectures and real world limitations. Unlike traditional surveys, this work emphasizes practical deployment challenges in resource constrained environments such as Bangladesh. Through a comparative analysis and contextual discussion, the study identifies key gaps in current systems, particularly in terms of connectivity dependence, computational requirements and lack of adaptability. To address these challenges, this paper highlights the importance of lightweight, context aware and connectivity resilient NLP based agentic systems.

References

Manish Adawadkar. 2025. The evolution of generative and agentic ai: From rule-based systems to autonomous intelligence. *American Journal of Technology Advancement Vol*, 2(11).

Mousa Ahmad Al-Bashrawi, Mohammed A Al-Sharafi, Ibrahim A Elgendy, Mohamed YI Helal, Madhan Karthikeyan Anbalagan, Inyoung Chae, and Yogesh K Dwivedi. 2026. Agentic ai systems and the future of entrepreneurship: a perspective on co-agency, innovation, and ecosystem transformation. *International Entrepreneurship and Management Journal*, 22(1):27.

Chimaobi Amadi and Adegboyega Ojo. 2025. Building trustworthy ai in healthcare. *IEEE Access*, 14:1182–1212.

Savinu Aththanayake, Chemini Mallikarachchi, Sajeev Kugarajah, and Janeesha Wickramasinghe. 2025. Resqconnect: Agentic ai for human-centered disaster response systems. In *2025 9th SLAAI International Conference on Artificial Intelligence (SLAAI-ICAI)*, pages 1–6. IEEE.

Ling Bai, Zirui Lin, Zheng Liu, Pengcheng Xi, Gaozhi George Xiao, and Shun Okuhara. 2025. Llama3-sise: An llm-based pipeline for preclinical staging of alzheimer's disease. In *2025 IEEE International Conference on Agentic AI (ICA)*, pages 232–235. IEEE.

Sarfraz Brohi, Qurat-ul-ain Mastoi, NZ Jhanjhi, and Thulasyammal Ramiah Pillai. 2025. A research landscape of agentic ai and large language models: Applications, challenges and future directions. *Algorithms*, 18(8):499.

Tom Brown, Benjamin Mann, and 1 others. 2020. Language models are few-shot learners. *Advances in Neural Information Processing Systems*.

Ritu Chauhan, Aarushi Mishra, Eiad Yafi, and Megat F Zuhairi. 2026. An agentic ai framework for adaptive symptom clarification and knowledge-augmented disease prediction in healthcare applications. In *2026 20th International Conference on Ubiquitous Information Management and Communication (IMCOM)*, pages 1–5. IEEE.

Arunraju Chinnaraju. 2025. Ai-powered consumer segmentation and targeting: A theoretical framework for precision marketing by autonomous (agentic) ai. *Int. J. Sci. Res. Arch*, 14:401–424.

Peerawat Chomphooyod, Lattapon Jeerapradit, Atiwong Suchato, and Proadpran Punyabukkana. 2025. Multi-agentic ai for automatic course syllabus generation using langgraph. In *2025 10th International STEM Education Conference (iSTEM-Ed)*, pages 1–6.

Xin Jie Chua, Jeraelyn Ming Li Tan, Jia Xuan Tan, Soon Chang Poh, Yi Xian Goh, Debbie Hui Tian Choong, Foong Chee Mun, Sze Jue Yang, and Chee Seng Chan. 2025. Banking done right: Redefining retail banking with language-centric ai. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 646–658.

Om Dabral, Swayam Bansal, Mridul Maheshwari, Hardik Sharma, Jaspreet Singh, and Bagesh Kumar. 2025. Multimodal trait and emotion recognition via agentic ai: An end-to-end pipeline. In *Proceedings of the 1st International Workshop on Cognition-oriented Multimodal Affective and Empathetic Computing*, pages 39–46.

Shivani Dhiman, Anjali Thukral, and Punam Bedi. 2026. An agentic ai system for disease diagnosis with explanations. *Informatics and Health*, 3(1):32–40.

Youness Fahmi and Mohammed Bousmah. 2025. Designing an educational virtual assistant based on agentic ai for student diagnosis and individualized remediation. In *2025 IEEE 8th Congress on Information Science and Technology (CiSt)*, pages 406–415.

Zijian Fang, Chao Yu, Rongxin Yang, Caixin Cai, Yucong Zhang, and Yinqi Wei. 2025. Douagent: A novel hierarchical agent framework in imperfect-information games. In *2025 IEEE International Conference on Agentic AI (ICA)*, pages 104–109. IEEE.

Deepak Narayan Gadde, Keerthan Kopparam Radhakrishna, Vaisakh Naduvodi Viswambharan, Aman Kumar, Djones Lettnin, Wolfgang Kunz, and Sebastian Simon. 2025a. Hey ai, generate me a hardware code! agentic ai-based hardware design & verification. In *2025 38th SBC/SBMicro/IEEE Symposium on Integrated Circuits and Systems Design (SBCCI)*, pages 1–5. IEEE.

Deepak Narayan Gadde, Keerthan Kopparam Radhakrishna, Vaisakh Naduvodi Viswambharan, Aman Kumar, Djones Lettnin, Wolfgang Kunz, and Sebastian Simon. 2025b. Hey ai, generate me a hardware code! agentic ai-based hardware design verification. In *2025 38th SBC/SBMicro/IEEE Symposium on Integrated Circuits and Systems Design (SBCCI)*, pages 1–5.

Ryan George, Akshay Govind Srinivasan, Jayden Koshy Joe, Harshith MR, Hrushikesh Kant, Rahul Vimalkanth, Sudharshan Suresh, and 1 others. 2025. Enhancing financial rag with agentic ai and multi-hyde: A novel approach to knowledge retrieval and hallucination reduction. In *Proceedings of The 10th Workshop on Financial Technology and Natural Language Processing*, pages 19–32.

Amit Giloni, Chiara Picardi, Roy Betser, Shamik Bose, Aishvariya Priya Rathina Sabapathy, and Roman Vainshtein. 2025. Cair: Counterfactual-based agent influence ranker for agentic ai workflows. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 18951–18977.

Subhajit Haldar, Souvik Sengupta, and Asit Kumar Das. 2025. Rubra: An agentic ai system for automatic short answer grading using llms and rag. In *2025*

International Conference on Computing, Intelligence, and Application (CIACON)*, pages 1–6. IEEE.

Agus Hasan, Ege Kandemir, Dmitrij Mordasov, and Dong Trong Nguyen. 2026. Agentic ai in wind energy systems: Multi-agent architectures for optimization and resilience. *IEEE Access*, 14:9935–9951.

Jiazhen He, Helen Lai, Lakshidaa Saigiridharan, Gian Marco Ghiandoni, Kinga Jenei, Umur Gokalp, Ajsa Nukovic, Ola Engkvist, Jon Paul Janet, and Samuel Genheden. 2026. Democratising real-world drug discovery through agentic ai. *Drug Discovery Today*, page 104605.

Pooja Honna, Sujith Chand, Neha Dinesh Rangdal, Ayush Patravali, Rachana MS, and 1 others. 2025. Agentic ai approaches for indian medicinal plant knowledge systems. In *2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS)*, pages 1–6. IEEE.

Gopichand Kanumolu, Ashok Urlana, Vinayak Kumar Charaka, and Bala Mallikarjunarao Garlapati. 2025. Agent ideate: A framework for product idea generation from patents using agentic ai. In *Proceedings of the 2nd Workshop on Agent AI for Scenario Planning*, pages 50–57.

Nalan Karunanayake. 2025. Next-generation agentic ai for transforming healthcare. *Informatics and Health*, 2(2):73–83.

Alaa Khamis. 2026. Design and evaluation of an agentic ai framework for personalized umrah trip planning. *Arabian Journal for Science and Engineering*, pages 1–21.

Alfredo Madrid-García, Diego Benavent, and Beatriz Merino-Barbancho. 2025. From chat to act: large language model agents and agentic ai as the next frontier of ai in rheumatology. *EULAR Rheumatology Open*, 1(3):147–156.

Jaykumar Ambadas Maheshkar, Himaja Vankayala, Vamshi Krishna Jakkula, Leslie Daniel Raj, Pratik Khedekar, and Rohit Laheri. 2026. Agentic ai-powered autonomous software engineering framework for automated code generation and debugging.

Tomas Mikolov, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space. *arXiv preprint arXiv:1301.3781*.

Jason H Moore and Nicholas P Tatonetti. 2025. From prompt engineering to agent engineering: expanding the ai toolbox with autonomous agentic ai collaborators for biomedical discovery. *BioData Mining*, 18(1):78.

Lino Murali, Aleena Thankachan, and Daleesha M Viswanathan. 2025. Agentic ai for severity extraction in clinical notes: Enhancing disease diagnosis beyond rule-based and traditional ml models. In *2025 International Conference on Power, Instrumentation, Control, and Computing (PICC)*, pages 1–6. IEEE.

N Kishor Narang. 2025. Mentor's musings on "agentic ai & physical ai (the disruptive new avatars of llms) empowering a truly autonomous iot future". *IEEE Internet of Things Magazine*, 8(6):4–13.

Farhad Nooralahzadeh, Yi Zhang, Jonathan F¨urst, and Kurt Stockinger. 2025. Multi-modal data exploration via language agents. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, pages 795–813.

Collins P Obeng, Nethshan M Narasinghe, Ryan Striker, and Enrique Alvarez Vazquez. 2025. Security framework for agentic home ai in preventive healthcare: Cyber threats worth noting. In *2025 Cyber Awareness and Research Symposium (CARS)*, pages 1–5.

Organisation for Economic Co-operation and Development. 2005. *OECD SME and Entrepreneurship Outlook*. OECD Publishing.

Rakesh Ramakrishna Pai. 2025. Agentic ai implementation in healthcare insurance industry: A comprehensive framework for automated claims processing and risk assessment. In *2025 IEEE Madhya Pradesh Section Conference (MPCON)*, pages 812–817. IEEE.

Joon Sung Park, Joseph O'Brien, and 1 others. 2023. Generative agents: Interactive simulacra of human behavior. *Proceedings of the ACM Symposium on User Interface Software and Technology*.

Akshar Patel, Stanley Joseph, Caryl Bailey, Ashish Sakharpe, and Mallikarjuna Devarapalli. 2025. Transforming electrophysiology workflows with natural language processing and agentic ai. *Heart Rhythm O2*.

Aske Plaat, Max van Duijn, Niki Van Stein, Mike Preuss, Peter van der Putten, and Kees Joost Batenburg. 2025. Agentic large language models, a survey. *Journal of Artificial Intelligence Research*, 84.

Roger S. Pressman. 2010. *Software Engineering: A Practitioner's Approach*. McGraw-Hill.

Chevean Richards and Bo Yang. 2025. Agentic ai, automating plane crash insights, reporting and recommendation. In *2025 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)*, pages 1–9. IEEE.

Stuart J. Russell and Peter Norvig. 2021. *Artificial Intelligence: A Modern Approach*, 4 edition. Pearson.

Gianina Salomó-López, Cristóbal Alcázar, Roberto Barceló, Camilo Carvajal Reyes, Darinka Radovic, and Felipe Tobar. 2025. Using language models to detect and reduce gender bias in university forum messages. *IEEE Signal Processing Magazine*, 42(6):95–109.

Lijun Sun, Yijun Yang, Qiqi Duan, Yuhui Shi, Chao Lyu, Yu-Cheng Chang, and Chin-Teng Lin. 2025. Multi-agent coordination across diverse applications: A survey. *arXiv preprint arXiv:2502.14743*.

NL Padma Swati, S Vanshita Gupta, Navya Sri Duddela, and L Rama Parvathy. 2026. Agentic ai-driven autonomous decision support system for smart agriculture. *Scientific Reports*.

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lei Wang, Yuntian Ma, and 1 others. 2023. A survey on large language model based autonomous agents. *arXiv preprint arXiv:2308.11432*.

Taylor Webb, Shanka Subhra Mondal, and Ida Momennejad. 2025. A brain-inspired agentic architecture to improve planning with llms. *Nature Communications*, 16(1):8633.

Yusheng Xie, Junyi Zhang, Haoyang Zhu, and 1 others. 2024. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*.

Yuhao Yang, Jiabin Tang, Lianghao Xia, Xingchen Zou, Yuxuan Liang, and Chao Huang. 2025. GraphAgent: Agentic graph language assistant. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics.

Shunyu Yao, Jeffrey Zhao, and 1 others. 2023. React: Synergizing reasoning and acting in language models. *arXiv preprint arXiv:2210.03629*.

Manish Yerram, Karthik Panicker, Mahendra Koneru, and Jatin Chopra. 2025. Llm-driven smart city management: Deploying agentic ai models with autogen and crewai. In *2025 International Conference on Digital Innovations for Sustainable Solutions (ICDISS)*, pages 1–6. IEEE.

Wayne Xin Zhao, Kun Zhou, Junyi Zhang, Haoyang Zhu, Sha Yuan, Jinhao Xu, Ji-Rong Wen, and Jian-Yun Nie. 2024. A survey on large language model-based autonomous agents. *Frontiers of Computer Science*, 18(6):1–26.

A Appendix A: Paper Selection Methodology

This survey is built upon a carefully curated collection of approximately 40 research papers gathered from major academic platforms, including IEEE Xplore, ACL Anthology and Google Scholar. In contrast to traditional surveys that rely heavily on existing review papers, this study adopts a more deliberate and exploratory approach.

Due to the relatively recent emergence of NLP driven agentic AI in software engineering, there exists a noticeable gap in unified survey literature directly addressing this intersection. As a result, this work intentionally avoids over reliance on secondary surveys and instead focuses on primary research contributions from multiple related domains.

The selection process follows a balanced and inclusive strategy. Papers are chosen with equal consideration from key research areas, including agentic AI architectures, natural language processing systems, multi agent collaboration frameworks and software engineering applications. This equal distribution ensures that no single domain dominates the narrative, allowing the study to form a holistic and unbiased understanding of the field.

Selection criteria include:

- Publications from well established and credible venues (e.g., IEEE, ACL)
- Relevance to agentic AI, NLP or software engineering workflows
- Contribution to system design, architecture or real world application
- Inclusion of both single agent and multi agent paradigms

Additionally, preference is given to recent works to reflect the latest advancements in the field. By weaving together insights from multiple domains, this study does not merely summarize existing research, but critically reinterprets it through the lens of software engineering in resource constrained environments.

This cross domain synthesis ultimately allows the paper to bridge fragmented research efforts and uncover practical insights that would remain hidden within isolated studies.